Hamming Distance

Time limit: 1.00 s **Memory limit:** 512 MB

The Hamming distance between two strings a and b of equal length is the number of positions where the strings differ.

You are given n bit strings, each of length k and your task is to calculate the minimum Hamming distance between two strings.

Input

The first input line has two integers n and k: the number of bit strings and their length.

Then there are n lines each consisting of one bit string of length k.

Output

Print the minimum Hamming distance between two strings.

Constraints

- $2 \le n \le 2 \cdot 10^4$
- $1 \le k \le 30$

Example

Input	Output
5 6 110111 001000 100001 101000 101110	1

Explanation: The strings `101000` and `001000` differ only at the first position.